

1 Radicals and Complex Numbers

Radical Expression is the form $\sqrt[n]{a}$. The $\sqrt[n]{}$ is a radical sign and the expression under it is the *radicand*, while the n is called the *index*.

$$\boxed{\text{If } x = \sqrt[n]{a}, \text{ then } x^n = a}$$

1.1 Adding and Subtracting Radical Expressions

Radical expressions are called **like radical expressions** if the indexes and radicands are the same.

Example: $3\sqrt{7} + 5\sqrt{7} = 8\sqrt{7}$

1.2 Multiplying Radical Expressions

To multiply we simply distribute each one then add them.

1.3 Dividing Radical Expressions

To divide we simply apply the sqrt to both top and bottom.

1.4 Rational Exponents

if n is a natural number greater than 1, and b is any nonnegative real number, then:

$$b^{1/n} = \sqrt[n]{b}$$

1.5 Complex Numbers

Imaginary values and pure imaginary numbers. The expression $\sqrt{-1}$ has no real answer. The symbol for it is i because its called an imaginary value. Any expression that is a product of a real number with i is called a pure imaginary number.

$$i = \sqrt{-1}, i^2 = -1$$